Project 2 Proposal: Text-to-Image Generation with CLIP and Stable Diffusion

IE 7615 Deep Learning Architectures · Project 2, Milestone 1 · Group 02: Abdellah Faleh, Jin-Woo Hong, Samuel Tong, Tristan Lyons · October 20, 2026

Overview

We will build a system that turns a short text description into a picture. You type a caption like "A dog runs on the beach", and the system creates a 512×512 image of it. The system has two main parts. A **text encoder** (CLIP) reads the caption and turns it into numbers. An **image generator** (Stable Diffusion 1.5) uses those numbers to create the picture.

This project is the opposite of Project 1. Project 1 looked at a picture and predicted a label. Project 2 starts from text and creates a new picture.

Dataset

We use **2,000 photos with captions** from the **Flickr30k** dataset (about 31,000 everyday photos, each with 5 captions written by people). We split them into 1,600 for training, 200 for validation and 200 for final testing. We crop and resize every photo to 512×512 pixels, clean the captions, and remove bad or repeated pairs (details in the dataset memo). Why Flickr30k:

- **Good captions.** People wrote them while looking at the photos, so text and image match well.
- **Right size.** 2,000 pairs are quick to download and prepare on free Google Colab.
- **Allowed for school projects.** The dataset is shared for research and education.
- **A real challenge.** Most photos show people. Image generators often struggle with faces, hands and groups, so we will learn a lot from the mistakes.

Model plan

We use **Stable Diffusion 1.5** together with the **CLIP text encoder**. CLIP turns the caption into 77 small vectors of numbers. The image generator (U-Net) starts from random noise and removes it step by step; at each step it looks at the caption's numbers, so the picture slowly becomes what the caption describes. A decoder then makes the final 512×512 image.

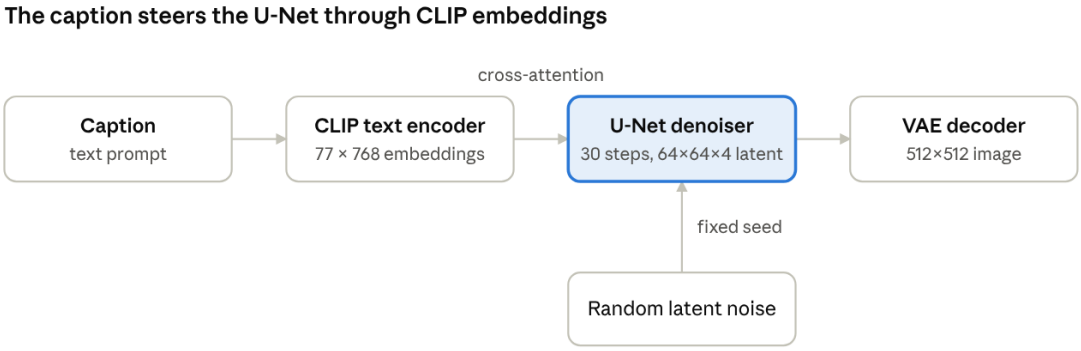

Figure 1. How the caption guides image generation in Stable Diffusion 1.5.

- **Milestone 1 (starting point):** use the ready-made model with normal settings (30 steps, guidance 7.5) on 15 test captions, and put each image next to the real photo.
- **Milestone 2 (the text part):** pass the caption's numbers into the generator with our own code; compare two text encoders, CLIP and BERT; test the guidance scale (how strongly the image follows the caption), the number of steps, and the noise-removal method (scheduler).
- **Milestone 3 (measuring quality):** measure results with FID, Inception Score and CLIP score, and test how much each setting changes them. If we have enough computer time, we also fine-tune on our training photos.

How we will measure results

We check three things: do the images look real, do they match the caption, and are they different from each other? We always use the same 200 test captions.

Score	What it tells us	Good result
FID	How close our images are to real photos	lower
Inception Score	How sharp and varied the images are	higher
CLIP score	How well each image matches its caption	higher
Seed similarity	Whether one caption always gives the same image (mode collapse)	not too high
Looking at images	Mistakes the scores miss: strange faces, hands, wrong number of people	—

Team roles

Each person leads one area. Everyone commits to the shared GitHub repository and checks one teammate's work each milestone. Tristan Lyons is the main contact with the teaching staff.

Role	Person	Main tasks
Technical lead	Abdellah Faleh	Data, CLIP and image-generation notebooks; connects the text encoder in Milestone 2
Experiments	Jin-Woo Hong	Runs the setting tests in Milestones 2 and 3
Evaluation	Samuel Tong	FID, Inception Score and CLIP score; result tables and charts
Documentation	Tristan Lyons	README, memos and report; repository organization; Canvas submission

Risks and how we handle them

Risk	What we will do
Limited GPU time: free Colab GPUs disconnect and have limited memory	Use faster settings, save every image to Google Drive as it is made, fine-tune only if time allows
Dataset license: the photos belong to their Flickr owners	Keep the 2,000 dataset photos out of GitHub; our notebook downloads them again; use them only for this course
Model license: Stable Diffusion uses CreativeML OpenRAIL-M	Keep the safety filter on; course use only; mention the license in the report
Unreliable scores: FID changes a lot with few images	Same number of images and same seeds for every test; also report CLIP score
Bias: some captions guess race or gender; the model learned internet biases	Point out these cases in our image review and discuss them in the report
BERT may not work: the generator was trained only with CLIP	A failed test is still a result; we will explain why it failed

Timeline

Milestone 1 (end of Module 7): dataset, CLIP pipeline, first images, proposal, repository · **Milestone 2** (end of Module 10): text encoder tests and settings · **Milestone 3** (end of Module 12): scores and report draft · **Milestone 4** (end of Module 13): final report and repository · **Presentation** (Module 14)